Stress-Testing a One-Dimensional Truth Axis

Seven Falsification Attempts on an Unsupervised Geometric Probe in Qwen2.5-1.5B

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Companion robustness study to *"How Much of Truth Fits on a Single Axis?"*

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Abstract

A companion paper showed that the truth value of a factual statement in Qwen2.5-1.5B can be read from an *unsupervised* geometric axis – the leading singular direction of paired true/false hidden-state differences – at cost $O(d)$, with no training and no polarity labels, and argued that on facts the model knows this axis is effectively one-dimensional. A single positive result invites the obvious objection: is the one-dimensionality real, or merely the easiest thing to see? This paper answers by attack. We subject the one-dimensional claim to seven independent falsification attempts, each designed to expose a richer structure if one exists: a complex/phase second dimension, a global depth signature, a dynamic "butterfly" propagation, a depth-department decomposition, an energy (magnitude) coordinate, a phase (argument) coordinate, and a three-dimensional class-clustering. Every attempt is run with the same shield – held-out cross-validation over pairs and a label-permutation null – or, for the purely observational tests, with an explicit separation measure. None overturns the claim. The depth signature is statistically real but never practically beats a single layer (+0.015 to +0.019 AUC); the depth-department curve is a smooth ramp, not steps; the energy and phase coordinates separate at chance ($|z|$ AUC ≈ 0.55, θ AUC $\approx$ 0.50–0.58); and in three dimensions true and false do not form distinct clumps (centroid-separation ratio 0.48–0.55, i.e. overlapping). What survives is precise: truth in this model is one-dimensional, carried by the position axis, concentrated by depth, and its separability scales with how well the model knows the fact (CounterFact 0.795 vs. TruthfulQA 0.656). We read a result that resists seven independent falsifications as strengthened, not weakened.

Reproducibility note (read first). The controlled experiments below are reproduced by the single script `truth_probe.py` (companion paper); the *observational* tests (attempts 5–7: the energy, phase, and three-dimensional class-clustering views) use a separate, purely visual script, `canvas.py`, which is posted alongside this paper. `canvas.py` renders – it does not classify or gate – so it is kept separate from the experimental harness. The exact commands for both scripts are listed in Section 6. A representative invocation for the visual tests is

```
python canvas.py --dataset counterfact --layer 16 --out counterfact
python canvas.py --dataset truthfulqa --layer 16 --out truthfulqa
```

1 Scope and method

This is a robustness study, not a new positive result. The claim under test, established in the companion paper, is:

Table 1: Seven attempts to falsify the one-dimensional truth axis. "Exp." = controlled experiment (held-out CV + permutation null); "Obs." = visual observation (descriptive separation statistic). None overturns the claim.

#	Richer structure hypothesized	Type	Result (real numbers)	Verdict
1	A complex/phase second dimension (truth as rotation)	Exp.	PHASE AUC 0.928 vs. 1D 0.938 (clean); reduces to the real-axis sign flip	falsified
2	Truth is global across depth (fixed-axis signature)	Exp.	global 0.923 vs. local 0.938 (-0.015), p=0.0099; persistence gap +0.04	falsified
3	Truth is dynamic – a "butterfly" in the unsaturated shade, on hard data	Exp.	global 0.642 vs. local 0.627 (+0.015), p=0.0099; commitment true 2.83→5.91 vs. false 2.65→5.86	falsified
4	Truth is distributed across layer "departments"	Exp.	diagnostic curve is a smooth ramp (no steps); global 0.646 vs. local 0.627 (+0.019), p=0.0099	falsified
5	The argument θ (phase) carries truth	Obs.	θ AUC 0.504 / 0.584 / 0.556 (CF / TQA / mix)	falsified
6	The energy $\|z\|$ (commitment) carries truth	Obs.	$\|z\|$ AUC 0.560 / 0.558 / 0.543	falsified
7	True/false form distinct 3D clumps	Obs.	centroid ratio 0.55 / 0.51 / 0.48 (< 1, overlapping); Re alone 0.795 / 0.656	falsified

In Qwen2.5-1.5B, the truth value of a factual statement is carried by a single (one-dimensional) direction – the position axis obtained without supervision from the SVD of paired true/false differences – on facts the model knows; the second and higher dimensions add no separation.

A claim of one-dimensionality is falsifiable in a specific way: exhibit a second, third, or depth-distributed coordinate that separates true from false better than the single axis. We construct seven such coordinates and test each.

Two kinds of test, kept distinct. We separate *controlled experiments* from *visual observations*, and label every result as one or the other. Controlled experiments (attempts 1–4) produce a held-out AUC under k-fold cross-validation *over pairs* (a pair is never split across folds) and a label-permutation null (true/false swapped within pairs) that quantifies selection bias; a result is meaningful only if it clears that null. Visual observations (attempts 5–7) do not classify; they project the states and measure a single descriptive statistic – the per-coordinate separation AUC, or the centroid-separation ratio – so the conclusion follows from a number attached to a picture, not from a fitted model. The model is frozen Qwen2.5-1.5B (28 blocks, 29 hidden-state levels, $d = 1536$), last-token pooling, peak layer 16 unless stated.

2 Seven falsification attempts

Table 1 summarizes all seven with their real numbers; the prose below gives the reasoning for each.

1 – The complex plane. The original framing promoted the second SVD direction to an imaginary axis and read truth as a phase rotation. If real, a phase- or magnitude-based score should beat the real axis. It does not: on clean minimal pairs the phase score reaches AUC 0.928 against 0.938 for the real axis, and a controlled synthetic check shows that with a pure sign flip and *zero* genuine rotation an angular score still separates, because a state at $\mathrm{Re} < 0$ has phase $\approx \pi$ by definition. The "rotation" is the sign of the real axis re-expressed in polar coordinates. The complex framing is decorative; we retract it.

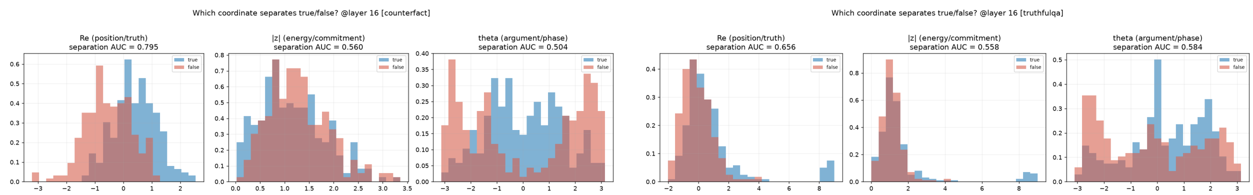
2 – A global depth signature. If truth flows through the network, the full trajectory $[\Delta_0, \ldots, \Delta_{28}]$ of the signed deviation on a fixed axis should beat the single best layer. On clean pairs it does not: the depth-signature classifier reaches held-out AUC 0.923 against 0.938 for the single layer (-0.015), and although the signature is real (permutation $p = 0.0099$), it carries no advantage. The crossing-persistence gap between false and true is $+0.04$ – flat. No propagation.

3 – A dynamic butterfly in the shade. The signed deviation saturates (σ compresses large values to $\pm\frac{1}{2}$), so a propagating *uncertainty* could be invisible to it. We re-ran on hard data (CounterFact/TruthfulQA mix) with the unsaturated features $[\mathrm{Re}, |z|]$ – the "shade." The depth signature now beats the single layer, but by $+0.015$ (0.642 vs. 0.627, $p = 0.0099$): real and marginal. The decisive descriptive measure kills the butterfly: the commitment magnitude grows identically for both classes (true $2.83 \to 5.91$, false $2.65 \to 5.86$); false statements do not hesitate more than true ones, and there is no propagating divergence.

4 – Layer departments. If different layer bands perform different jobs (surface, semantics, logic), a single semantic axis re-read everywhere wastes most layers; giving each layer its *own* native probe should recover a distributed signal, and the per-layer separation curve should show steps. It shows a smooth ramp instead – the per-layer native separation rises continuously to a single peak at layer 16, with no plateaus and no jumps. The per-layer signature beats the single layer by $+0.019$ (0.646 vs. 0.627, $p = 0.0099$) – again real and marginal. No departments; the native-axis "cure" improves the fixed axis but does not change the ceiling.

5–6 – Energy and argument. Decomposing each state into position (Re), energy ($|z|$), and argument (θ) and measuring how well each separates true from false (Fig. 1): only position carries signal. The energy separates at $|z|$ AUC 0.560/0.558/0.543 (CounterFact/TruthfulQA/mix) – true and false are equally committed. The argument separates at θ AUC 0.504/0.584/0.556 – essentially chance; what little it shows is the echo of the Re sign, not independent information. The phase, the very coordinate the complex framing made central, is empty.

7 – Three-dimensional clumping. If truth were a richer object, true and false should form distinct clusters in the three-dimensional (v_1, v_2, v_3) space. They do not (Fig. 2). The centroid-separation ratio – distance between the true and false centroids divided by the average cloud spread – is 0.55/0.51/0.48, all below 1: the clouds overlap. Crucially this low ratio is *not* evidence against the axis; it is the one-dimensional signal seen through two noise dimensions. Projected onto position alone, the same data separate at AUC 0.795 (CounterFact). Adding the orthogonal dimensions dilutes, it does not enrich.



(a) CounterFact: Re 0.795, $|z|$ 0.560, θ 0.504

(b) TruthfulQA: Re 0.656, $|z|$ 0.558, θ 0.584

Figure 1: Attempts 5–6 and knowledge scaling, layer 16. Per-coordinate separation AUC. On both datasets only the position axis (Re) separates true from false; energy ($|z|$) and argument (θ) are at chance. The position separation falls from 0.795 (CounterFact, often-known facts) to 0.656 (TruthfulQA, more ambiguous), while the one-dimensional structure is unchanged.

3 What survives

Three statements resist all seven attempts.

Truth is one-dimensional and lives on the position axis. Across every decomposition, only Re separates; energy, phase, the second and third SVD directions, and the full-depth trajectory add at most +0.02 AUC and usually nothing.

Separability scales with knowledge. On CounterFact, where the model often knows the fact, the position axis separates at AUC 0.795; on the more ambiguous TruthfulQA at 0.656; on the diffuse mix the centroids nearly coincide. The clearer the model's knowledge, the more one-dimensional and separable the truth representation.

On unknown facts, the classes show low separability. Where the model does not know the fact, true and false representations do not form distinct clusters but overlap in space – intuitively a dense, uniform distribution rather than two distinct clumps. We state this as *low class separability / distribution overlap* (centroid ratio < 1), the measured counterpart of the knowledge-dependent dimensionality reported in the companion paper.

4 The axis reads truth and confidence

The seven attempts establish that there is no second coordinate of truth to find. This has a constructive consequence that is easy to miss, so we state it plainly. The same unsupervised axis admits *two* readings, not one.

Its *sign* classifies a statement: Re > 0 is the true side, Re < 0 the false side. Its *separability* – how cleanly true and false split on a given set of facts – measures how well the model knows that material. The geometry that answers "is this statement true?" therefore also answers, indirectly, "does the model know this?", because a model that does not know produces overlapping rather than separated representations (Figs. 2, 3). These are not two axes; they are two readings of one axis – the direction of the arrow, and the sharpness of the arrow.

The reading is quantitative at the level of a distribution. The position-axis separation is 0.795 on CounterFact and 0.656 on TruthfulQA: by this measure the model "knows" the CounterFact material better than the TruthfulQA material, read off the geometry alone, with no external grading. We report this as a property of the dataset, not yet of the single fact. Calibrating a per-fact confidence – a local separability around one statement that predicts whether the model actually errs on it – is the natural next step, and requires a behavioral label (does the model answer the open question correctly?) that the minimal-pair design does not contain. We mark it as future work rather than a claim, to keep the boundary between what is measured and what is conjectured explicit.

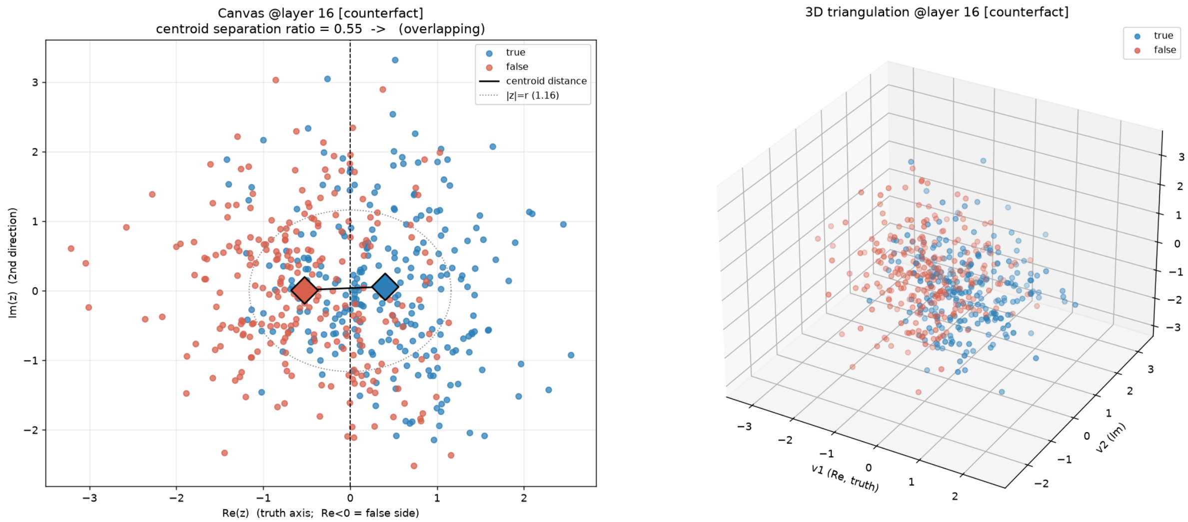


Figure 2: Attempt 7, CounterFact at layer 16. Left: the 2D plane; the true (blue) and false (red) centroids (diamonds) are close relative to the cloud spread (centroid-separation ratio 0.55, overlapping). Right: the three-dimensional (v_1, v_2, v_3) scatter shows a gradient along v_1 (position) but no distinct clumps. The separation is confined to the horizontal position axis; the extra dimensions are noise.

5 Why unknown facts overlap: the signature of superposition

It remains to interpret *why* the classes overlap on unknown facts rather than simply asking *that* they do. The overlap is not measurement noise; it is the observable signature of unresolved superposition, and the explanation follows from a capacity argument already implicit in the companion paper.

A projection onto a single direction returns a scalar: it collapses the d-dimensional state to one number and discards $d - 1$ dimensions. When the model knows a fact, its truth content has been allocated a clean, dominant direction of its own – the residual stream has, in effect, dedicated an axis to it – and the single projection recovers almost all of it (position AUC 0.795, near the full-probe ceiling). When the model does not know a fact, there is no dedicated direction: the weak truth-relevant signal remains distributed across many superposed features, and no single axis can pull it out of the tangle. Recovering it would require asking the question once per dictionary atom – the expansion that a sparse autoencoder performs at cost $O(d\,m)$ – which is a different operation from a single projection, not a cheaper version of it.

The consequence is exactly what the canvas shows. On known facts the truth occupies its own direction, the classes form two regions, and the centroid ratio is high. On unknown facts the truth is spread across the superposition, the classes overlap, and the centroid ratio falls below one. The dense, uniform overlap of true and false on unknown facts is therefore not a failure of the probe but a direct, visual readout of superposition that the probe structurally cannot resolve. The ≈ 0.20 probe gap reported in the companion paper and the low centroid ratio measured here are two views of the same quantity: the truth information that lives off-axis, in superposition, on facts the model does not know.

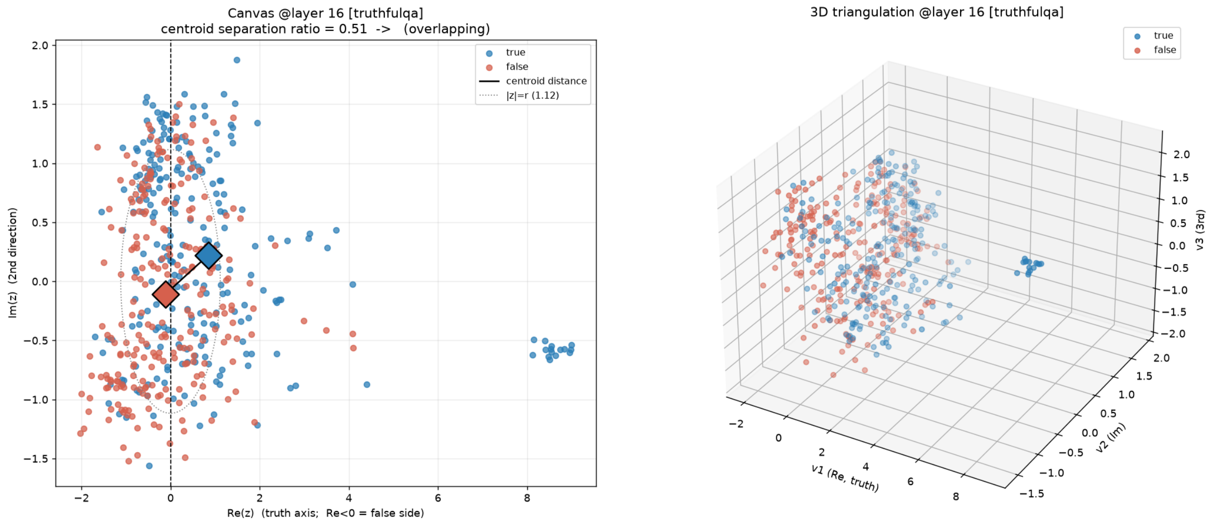


Figure 3: Attempt 7, TruthfulQA at layer 16. The same picture on more ambiguous facts: centroid-separation ratio 0.51 (overlapping), no three-dimensional clumps. Lower knowledge, lower separability, identical one-dimensional structure.

6 Reproducibility

Two scripts reproduce every number. `truth_probe.py` (companion paper) runs the controlled experiments; `canvas.py` (posted with this paper) renders the visual observations. Both load Qwen2.5-1.5B with safetensors only and load external datasets as Parquet only, never with remote code. A CUDA GPU is optional.

Controlled experiments (attempts 1–4).

```
# survivor + baseline (and attempt 1, the PHASE ablation)
python truth_probe.py signal --dataset builtin --with-2d --baseline --perm 200

# attempt 2: global depth signature (fixed axis), clean pairs
python truth_probe.py domino --dataset builtin --perm 100

# attempt 3: dynamic shade on hard data
python truth_probe.py domino --dataset mix --max-pairs 250 --signal-mag --perm 100 \
       --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
       --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490

# attempt 4: per-layer native axes + diagnostic ramp
python truth_probe.py domino --dataset mix --max-pairs 250 --per-layer-axis --perm 100 \
       --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
       --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Visual observations (attempts 5–7). The figures in this paper are the output of:

```
python canvas.py --dataset counterfact --layer 16 --out counterfact \
       --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

```
python canvas.py --dataset truthfulqa --layer 16 --out truthfulqa \
       --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Each run writes `<out>_coords.png` (per-coordinate separation), `<out>_plane.png` (the 2D plane with centroids and the separation ratio), and `<out>_3d.png` (the three-dimensional scatter). Upload the relevant PNGs to the Overleaf project so the figures render, or comment out the figure blocks.

7 Limitations and conclusion

These results are confined to a single, small model (Qwen2.5-1.5B); the entire battery should be re-run on a larger model and a different family before the one-dimensionality is treated as general – it is plausible that departments, dynamics, or a genuine second dimension emerge with scale, precisely because a 1.5B residual stream may be too compact to separate them. The datasets are small, and the "unknown-fact" interpretation of low separability is plausible but confounded with sentence-format effects, which is why mixing datasets must be avoided (it injects a spurious format axis). Within these limits, the conclusion is stable: a result that survives seven independent falsification attempts – spanning a second dimension, depth-global structure, dynamics, departments, energy, phase, and three-dimensional clustering – is not weakened by the absence of richer structure. It is a measurement of a boundary: in this model, truth is one-dimensional, on the position axis, on the facts it knows, and nowhere else.

References

[1] F. K. Vicidomini. *How Much of Truth Fits on a Single Axis? Knowledge-Dependent Dimensionality and Unsupervised Polarity Recovery in LLM Representations.* Companion paper, 2025.

[2] L. Bürger, F. A. Hamprecht, B. Nadler. *Truth is Universal: Robust Detection of Lies in LLMs.* arXiv:2407.12831, 2024 (NeurIPS 2024).

[3] S. Marks, M. Tegmark. *The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets.* arXiv:2310.06824, 2023 (COLM 2024).

[4] N. Elhage et al. *Toy Models of Superposition.* Transformer Circuits Thread, Anthropic, 2022 (arXiv:2209.10652).